Assessment - 6
25BCB0074

No Sql

1. Creating dbs:

```
test> use NewStudentDB
switched to db NewStudentDB
NewStudentDB> show dbs
25BCB0074   72.00 KiB
admin       40.00 KiB
config      96.00 KiB
local      212.00 KiB
```

2. Inserting

```
NewStudentDB> db.createCollection('student')
{ ok: 1 }
NewStudentDB> db.student.insertOne({Roll_no: "25BCB0074",name: "Tanvi", stream: "Cse"})
{
  acknowledged: true,
  insertedId: ObjectId('6ab5eae583f6276f4c5b4355')
}
NewStudentDB> db.student.insertMany([{Roll_no: "25BCB0054",name: "pqr", stream: "Cse"}, {Roll_no:"25BCB0073",name:"Abc",stream:"Cse"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab5eb3a83f6276f4c5b4356'),
    '1': ObjectId('6ab5eb3a83f6276f4c5b4357')
  }
}
```

3. Fetching

```
NewStudentDB> db.student.find()
[
  {
    _id: ObjectId('6ab5eae583f6276f4c5b4355'),
    Roll_no: '25BCB0074',
    name: 'Tanvi',
    stream: 'Cse'
  },
  {
    _id: ObjectId('6ab5eb3a83f6276f4c5b4356'),
    Roll_no: '25BCB0054',
    name: 'pqr',
    stream: 'Cse'
  },
  {
    _id: ObjectId('6ab5eb3a83f6276f4c5b4357'),
    Roll_no: '25BCB0073',
    name: 'Abc',
    stream: 'Cse'
  }
]
NewStudentDB>
```

4. Updating

```
NewStudentDB> db.student.updateOne({Roll_no: "25BCB0073"},{$set: {name:"edf"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

5. Delete

```
]
NewStudentDB> db.student.deleteOne({Roll_no:"25BCB0054"})
{ acknowledged: true, deletedCount: 1 }
NewStudentDB>
```

Frontend - Backend connectivity

```
CREATE TABLE DESTINATION (
ID NUMBER PRIMARY KEY,
NAME VARCHAR2(50),
DESCRIPTION VARCHAR2(100)


public class JDBCConnectionTest {
 public static void main(String[] args) {
     String url = "jdbc:oracle:thin:@localhost:1521:xe";
     String username = "system";
     String password = "oracle";

     try {
          Class.forName("oracle.jdbc.driver.OracleDriver");
          Connection con = DriverManager.getConnection(url, username, password);
          if (con != null) {
              System.out.println("Connected successfully!");
          } else {
              System.out.println("Connection failed!");
          }

          String createTableQuery = "CREATE TABLE DESTINATION ("
                  + "ID NUMBER PRIMARY KEY, "
                  + "NAME VARCHAR2(50), "
                  + "DESCRIPTION VARCHAR2(100))";
          Statement stmt = con.createStatement();
          int result = stmt.executeUpdate(createTableQuery);
          if (result == 0) {
              System.out.println("Table created successfully!");
          } else {
              System.out.println("Table creation failed!");
          }
          stmt.close();
          con.close();
     } catch (Exception e) {
 }    System.out.println("Error: " + e.getMessage());
}
```

```
import javax.swing.*;
import java.awt.*;
import java.sql.*;

public class OracleDBSwingApp extends JFrame {
    JTextField idField, nameField, emailField;
    JButton insertButton, deleteButton, updateButton;
    Connection conn;

    public OracleDBSwingApp() {
        setTitle("Oracle DB CRUD Application");
        setSize(400, 350);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLayout(new GridLayout(5, 2, 5, 5));

        add(new JLabel("ID:")); idField  = new JTextField(); add(idField);
        add(new JLabel("Name:")); nameField = new JTextField(); add(nameField);
        add(new JLabel("Email:")); emailField = new JTextField(); add(emailField);

        insertButton  = new JButton("Insert");
        deleteButton  = new JButton("Delete");
        updateButton  = new JButton("Update");

        add(insertButton); add(deleteButton); add(updateButton);

        connectToDatabase();

        insertButton.addActionListener(e -> insertRecord());
        deleteButton.addActionListener(e -> deleteRecord());
        updateButton.addActionListener(e -> updateRecord());
    }
    void connectToDatabase() {
        try {
            Class.forName("oracle.jdbc.driver.OracleDriver");
            conn = DriverManager.getConnection("jdbc:oracle:thin:@localhost:1521:XE", "system", "123");
            JOptionPane.showMessageDialog(this, "Connected to Oracle DB successfully!");
        } catch (Exception ex) {
            ex.printStackTrace();
        }
    }
    void insertRecord() {
        String query = "INSERT INTO users (id, name, email) VALUES (?, ?, ?)";
        try (PreparedStatement pst = conn.prepareStatement(query)) {
            pst.setInt(1, Integer.parseInt(idField.getText().trim()))>
            pst.setString(2, nameField.getText().trim());
            pst.setString(3, emailField.getText().trim());
            pst.executeUpdate();
        } catch (Exception ex) {sageDialog(this, "Record successfully inserted into Backend!");
        }    JOptionPane.showMessageDialog(this, "Insert Failed: " + ex.getMessage());
    }
    // ... deleteRecord() and updateRecord() implementations continue here ...
    public static void main(String[] args) {
```

```
void connectToDatabase() {
try {
       Class.forName("oracle.jdbc.driver.OracleDriver");
       conn = DriverManager.getConnection("jdbc:oracle:thin:@localhost:1521:XE",
           "system", "123");
       JOptionPane.showMessageDialog(this, "Connected to Oracle DB successfully!");
} catch (Exception ex) {
       ex.printStackTrace();
}


void insertRecord() {
String query = "INSERT INTO users (id, name, email) VALUES (?, ?, ?)";
try (PreparedStatement pst = conn.prepareStatement(query)) {
       pst.setInt(1, Integer.parseInt(idField.getText().trim());
       pst.setString(2, nameField.getText().trim());
       pst.setString(3, emailField.getText().trim());
       pst.executeUpdate();
       JOptionPane.showMessageDialog(this, "Record successfully inserted into Backend!");
} catch (Exception ex) {
       JOptionPane.showMessageDialog(this, "Insert Failed: " + ex.getMessage());
}


SELECT * FROM users;
```